MANIPAL ACADEMY of HIGHER EDUCATION
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Question Paper - Report

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SONTAKKE MANGALAM ANAND .
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Question Paper

Back

MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech III Semester Mid Semester Examination September 2023

DATA STRUCTURES [DSE 2122]

Marks: 30 **Duration: 120 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) Consider the given infix expression**. K+L-M*N+(O^P)*W/U/B*T+Q**

During the conversion of the above expression into Polish notation, how many arithmetic operator symbols will be available **on the stack** (excluding parenthesis) after reading third + in the given expression (0.5)

4 1 3 2

2) Consider the given infix expression. **K+L-M*N+(O^P)*W/U/B*T+Q**

During the conversion of the above expression into Polish notation, how many arithmetic operator symbols will be available in the **resultant Polish form** after reading **-** in the given expression (0.5)

6 5 7 8

3) A _____traverses the data elements sequentially, in which only one data element can directly be reached.

(0.5)

Non Linear Data Structures | Linear Data Structures | Abstract Data type | Compound data structures

4) In which of the following data structures relationship of adjacency is not maintained between the data elements.

(0.5)

Arrays Stack Tree Queue

5) If the given upper triangular matrix is mapped to 1D array, then identify the correct formula to retrieve the value 6 from 1D array (0.5)

[

5 3 8
0 1 7
0 0 6

]

A[2*1+2-1*(1+1)/2] A[(3*1+1)/2+2] A[(3*2)+2-(2*(2+1)/2)] A[(3*1)*2-(1+(1+1)/2)]

6) What is the output for the following code segment? (0.5)

```
int main() { int a=10;// address of variables (a=1000, d1=1500, d2=2000)
int *d1,**d2;
d1=&a;
d2=&d1;
cout< < a;
cout< < *(&a);
cout< < **d2;

return 0; }
```

10, 1000, 1000 10,10,10 10, 1000, 10 10,1000, 2000

7) What will be the output of the following code segment? (0.5)

```
int main()
{ int x[3]={8,3,2};
int *p;
for(p=&x[1];p< =&x[2];p++)
cout< < *p;

return 0; }
```

8,3 3,2 8,3,2 8,2

8) The following postfix expression with single digit operands is evaluated using stack. (0.5)

8 2 3 ^ / 2 3 * + 5 1 * - Note that ^ is the exponentiation operator. The top two element of the stack after the first * is evaluated as _____ (Note: The first element in the given options represent the top most element in stack)

1,6 6,1 3,2 2,3

9) What will be the output of the following code segment? (0.5)

```
int main()
{ int x[25];
x[0]=10;
x[24]=10;
```

cout< < *x < < endl;
cout< < *(x+24)+*(x+0) < < endl;

return 0; }

10,20 20, 10 10,10 20, 20

10) Consider the following operation performed on anempty stack.

Push(1); Pop();Push(2); Push(3);Pop(); Push(4);Pop(); Pop(); Push(5); a=pop()

Consider the following operation performedon anempty queue.

Enqueue(2);Enqueue(3); Dequeue();Enqueue(4);Enqueue(5);b=Dequeue(); (0.5)

The value of **a*b=**_____

15 25 10 360

DESCRIPTIVE

Answer all the questions.

11) Given the algorithm fun(n), draw the recursion tree for n = 3.

ALGORITHM fun(n)

// Input : A non-negative integer n

if n = 0

return 1 (2)

else

return fun(n-1) + fun(n-1)

.

12) A) Write the statements to increment the rear and front of a circular queue (maximum size is C)?

B) Consider the given Queue elements as **4,5,6,7,8** of size 5. Explain the possible methods to delete the element 4 from the given queue. (3)

.

13) Check the given expression for parenthesis matching using **position** of brackets to validate the given expression.

(((a+b)*c+d-e)/(f+g)-(h+j)*(k-l)))/(m-n))

(3)

Write all the intermediate steps in detail along with the content of stack . Assume starting index is 0 .

14) (3)

Draw the Recursive Tree for the Towers of Hanoi problem where the number of the disks is four. The function call is **towers (3, Z,Y,X)** where 3 =no. of disks, Z=Source, Y=destination, X=Auxiliary at the initial step. Write all the moves from the constructed tree and verify the total moves using the appropriate formula .

15) Consider the given ordered singly linked list ={25,30,45,60,75}. Write the steps to be followed for insertion of element 32 in the given list. Also, write the steps to perform the deletion of element 60. (3)

16) Write the algorithm to perform enqueue and dequeue operation in standard queue. Use all the necessary functions in it. (3)

17) Write the polish notation form of the given infix expression using **standard** algorithm.

(((A+B)*C+D-E)/(F+G)-(H+J)*(K-L))/(M-N)

Write all the intermediate steps and show the stack content and respective resultant expression at each step . (4)

18) Perform the reverse Polish evaluation for the given expression:

AB^C*I^DEZ-*M/N+

Show all the intermediate steps and stack content at each step. Also mention the data structure used for this operation . (4)

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^ Top